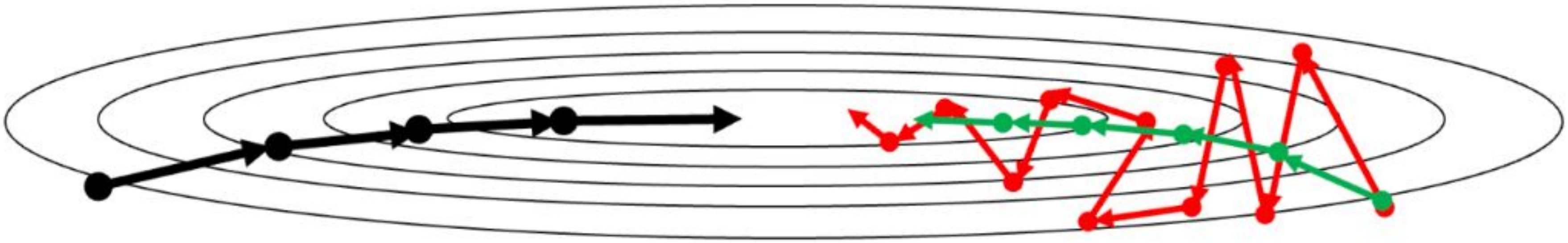
Introduction to Neural Networks

Seminar - Week 4 - Optimizers

Assignment 3 Goals

- Implement optimizers
  - SGD with Momentum
  - RMSProp
  - Adam

Gradient Descent - Momentum



Gradient Descent

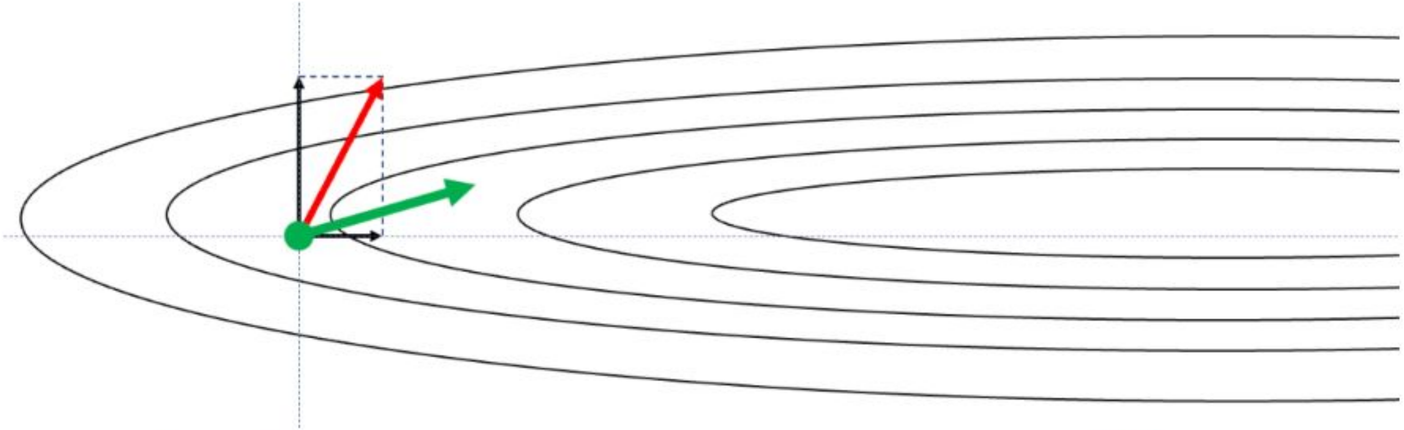
$$W := W - \alpha dW$$
$$b := b - \alpha db$$

Gradient Descent with Momentum

$$V_{dW} := (1 - \beta)dW + \beta V_{dW}$$
$$V_{db} := (1 - \beta)db + \beta V_{db}$$

$$W := W - \alpha V_{dW}$$
$$b := b - \alpha V_{db}$$

RMSProp



$$S_{dW} := (1-\beta)dW^2 + \beta S_{dW}$$

$$S_{db} := (1-\beta)db^2 + \beta S_{db}$$

$$W := W - \alpha \frac{dW}{\sqrt{S_{dW}}}$$

$$b := b - \alpha \frac{db}{\sqrt{S_{db}}}$$

Adam

$$V_{dW} := (1-\beta_1)dW + \beta_1 V_{dW}$$

$$S_{dW} := (1-\beta_2)dW^2 + \beta_2 S_{dW}$$

$$\widehat{V}_{dW} := \frac{V_{dW}}{1-\beta_1^t}$$

$$\widehat{S}_{dW} := \frac{S_{dW}}{1-\beta_2^t}$$

$$W := W - \alpha \frac{\widehat{V}_{dW}}{\sqrt{\widehat{S}_{dW}} + \epsilon}$$

Small value
for numerical stability

Kingma, Diederik P., and Jimmy Ba. "Adam: A method for stochastic optimization." arXiv preprint arXiv:1412.6980 (2014).

Week 4 - TASK

- Implement optimizers
  - SGD with Momentum
  - RMSProp
  - Adam

https://github.com/vgg-fiit/NSIETE_2026

Project 1 (8b)

- MLP
  - Paper presentation
  - Data analysis
  - Data preprocessing and normalization
  - Data split
  - Configuration
  - Experiment tracking
  - Experiments - meaningful based on the results of previous experiments
    - Including improvement techniques (e.g. Dropout, Normalization layers, Skip Connections, Bottleneck Layers, …)
  - Results and evaluation metrics
  - Clear code and documentation/comments
  - Final presentation of projects
  - Effort on consultations

Project 1

- Consultation 1 - minimal requirements
    - Select team
    - Select task (maximum 2 teams per task, first come, first serve)
    - Select paper
    - Download and examine dataset
    - Exploratory data analysis (example)
    - Select evaluation metrics
    - Check THIS and THIS video
    - Prepare presentation about the selected paper

Project 1 - List of topics

Credit Risk Assessment

- Binary Classification
- Dataset - https://api.openml.org/d/31

Speed Dating

- Binary Classification
- Dataset - https://api.openml.org/d/40536

Diabetes 130 US

- Binary Classification
- Dataset - https://api.openml.org/d/45069

Bioresponse

- Binary Classification
- Dataset - https://api.openml.org/d/4134

Magic Telescope

- Binary Classification
- Dataset - https://www.openml.org/data/download/54003/MagicTelescope.arff
  https://archive.ics.uci.edu/dataset/159/magic+gamma+telescope
- paper - https://publikationen.bibliothek.kit.edu/270043064/3813660
  - paper about the creation of the data

Detroit

- Regression
- Dataset - https://api.openml.org/d/552

Houses

- Regression
- Dataset - https://api.openml.org/d/537
-

Steel Plates Fault

- Multiclass Classification/Binary classification
- Dataset - https://api.openml.org/d/1504

Project 1 - List of papers

- **Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift, 2015**
- **Dropout: A Simple Way to Prevent Neural Networks from Overfitting, 2014**
- **Deep Residual Learning for Image Recognition, 2015 98 (PRELU)**
- **Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification, 2015**
- **Adam: A Method for Stochastic Optimization, 2014**
- **Wu, Y. and He, K., 2018. Group normalization. In Proceedings of the European conference on computer vision (ECCV) (pp. 3-19).**
- **Exact solutions to the nonlinear dynamics of learning in deep linear neural networks - Saxe, A. et al. (2013).**
- **Understanding the difficulty of training deep feedforward neural networks - Glorot, X. & Bengio, Y. (2010).**